

9.24 Solution: (a) Inside the conductor, assuming the conductivity is σ , the Maxwell equations become

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0, \quad \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J} = \sigma \mathbf{E}.$$

The wave equation is then

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu\sigma \frac{\partial \mathbf{E}}{\partial t} = 0.$$

Assuming harmonic time dependence, we can see that the characteristic equation is given by

$$\mu\epsilon\omega^2 + i\mu\sigma\omega - k^2 = 0,$$

compared to the corresponding equation in free space, $\mu\epsilon\omega^2 - k^2 = 0$. Solution to the characteristic frequency in terms of k is

$$\omega = -i\frac{\sigma}{2\epsilon} \pm \sqrt{\frac{k^2}{\mu\epsilon} - \frac{\sigma^2}{4\epsilon^2}},$$

which always has a negative imaginary part.

The value of k can be determined in the same way as in Problem 9.22, where for the TE modes, it is the solutions to the equation $j_l(ka) = 0$, while for TM modes,

$$\left. \frac{d}{dx} [xj_l(x)] \right|_{x=ka} = 0.$$